Jeremy Burke

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Professional Summary

Applied AI and software engineer delivering defense and aerospace R&D software across SBIR/STTR programs. Experienced in computer vision, sensor fusion, simulation, RAG systems, LLM tooling, backend APIs, and technical automation. Builds practical software systems that connect AI models, engineering data, and usable interfaces across \$100K–\$1M+ efforts.

Experience

Computer Engineer — Beacon Industries Aug 2025 – Present
Newington, CT

- Led SBIR/STTR technical programs across DoD, NASA, and MDA efforts, delivering reports, software prototypes, simulations, and technical analyses valued at \$30K–\$250K+ independently
- Developed computer vision models, synthetic sensor pipelines, and fusion-validation workflows for perception and multi-spectral detection programs
- Engineered custom plugins and simulation tools for thermal, environmental, and multi-spectral sensor modeling from raw data streams
- Built AI-enabled internal tools using AWS, Weaviate, OpenAI APIs, and custom automation pipelines for technical analysis and decision support
- Presented technical work, R&D outcomes, and commercialization opportunities to government stakeholders

Selected R&D Projects

- **Multi-Spectral Passive Object Detection, Navy SBIR:** Built ML-based object-detection pipeline using synthetic and real multi-spectral data; developed 3D physics simulations and sensor plugins for environmental modeling and fusion testing.
- **Hypersonic Kill Vehicle Research, MDA SBIR:** Performed ANSYS simulations to evaluate high-speed system performance, design tradeoffs, and mission-relevant operating conditions.
- **Additive Manufacturing R&D, SBIR Phase II:** Delivered \$500K in technical milestones and coordinated external research collaboration.
- **Reverse Engineering, Defense System:** Leading vendor coordination and technical analysis for legacy electromechanical system reconstruction.

Personal Projects

- **SentinelGrid — Climate-Risk Telemetry Platform:** End-to-end IoT pipeline — C++ sensor-fleet simulator, MQTT, FastAPI, PostgreSQL/PostGIS, and Python anomaly-scoring worker (z-score + Isolation Forest, sensor-drift quarantine) behind a Next.js/Leaflet national ops console (incident triage, playback, forecasts); deterministic in-browser simulation of 4,174 nodes across 19 US regions with zero-storage 24-hour replay, overlaid on live public data (NEXRAD radar, ∼3,700 NWS/USGS stations, active storm warnings); 90+ tests, 8 CI jobs.
- **ProjectPulse:** Production-style project-management platform — ASP.NET Core 8 Clean Architecture API with a React 19 dashboard; drag-and-drop Kanban with domain-enforced transitions, isolated demo workspaces, role-based permissions, rate limiting, and audit logging (xUnit/Vitest, Docker Compose, GitHub Actions CI).
- **Open Computer Vision Detection Dashboard:** Next.js/TypeScript + Cloud Run FastAPI dashboard serving 7 detectors across YOLO v8–v12, RT-DETR, and open-vocabulary YOLO-World; upload, webcam, and batch inference with annotated image, JSON, and CSV outputs.
- **Technical Paper AI Search Assistant:** Source-grounded RAG assistant on Cloudflare Workers/D1 — hybrid semantic + BM25 retrieval, in-browser PDF search (Transformers.js), exact-passage citations, and a fail-closed cost quota.

Education

University of Connecticut Aug 2019 – Dec 2023
B.S. Computer Science & Engineering, Minor Mathematics — GPA: 3.3 Storrs, CT

Technical Skills

- **Programming:** Python, TypeScript, JavaScript, C, C++, C#, Java, SQL
- **AI/ML:** YOLOv8, OpenCV, RAG, Sentence Transformers, LLM Systems, Multimodal Fusion
- **Web/Backend:** Next.js, React, FastAPI, ASP.NET Core 8, EF Core, REST APIs, Swagger/OpenAPI
- **Data/Search:** Chroma, BM25, Weaviate, PyMuPDF, Vector Search, Hybrid Retrieval
- **Simulation:** Gazebo, ANSYS Workbench, MATLAB, Magics, Physics-Based Modeling
- **Cloud/Tools:** AWS, Google Cloud Run, Docker, GitHub Actions, Firebase, OpenAI API, Ollama, Linux, Git, LaTeX